

Temporal Analysis of Urban Vegetation Changes Using Satellite Imagery and Spectral Indices

Ali Ömer Çetinkaya
2022401189

Devrim Deniz Yılmaz
2022401219

Problem Statement

Rapid urbanization and climate change continue to reshape the structure of major cities. As forests, parks, and other green zones shrink, the consequences include worsening air quality, stronger urban heat island effects, and declining ecological stability. Our project focuses on creating a practical system that automatically quantifies how the vegetation of a chosen city has changed over time. Using raw satellite data and applying image processing methods from our course, we aim to measure long-term trends in green area distribution and connect them to environmental drivers.

Introduction

Remote sensing is one of the most effective ways to monitor environmental changes over large regions. However, raw satellite images are rarely ready for direct analysis. They often suffer from three major issues:

- atmospheric haze that reduces contrast,
- uneven illumination due to mountains, shadows, or time of day,
- sensor noise that affects pixel reliability.

To obtain accurate vegetation indices, we first need to correct these distortions. In this project, we will work with Google Earth Engine Level-1 satellite data and build a restoration pipeline using:

- Dark Object Subtraction (to remove haze),
- Homomorphic Filtering (to correct illumination),
- Adaptive Noise Reduction (to reduce sensor noise while preserving edges).

After restoration, we compute spectral indicators such as NDVI, NDMI, and the HSI Hue Angle. These features allow us to perform a meaningful segmentation with K-means clustering.

Methodology

1. Data Acquisition and Pre-processing

We first select satellite images from Google Earth Engine using the `CLOUD_COVER` metadata. Since we use Level-1 data, this is our primary tool for filtering out cloudy scenes. After downloading the Red, NIR, and SWIR bands, we visualize them to ensure that the dataset is suitable for further processing.

2. Atmospheric Correction (Dark Object Subtraction)

To remove atmospheric haze, we use the degradation model described in Gonzalez & Woods:

$$g(x, y) = f(x, y) + \eta(x, y)$$

where $g(x, y)$ is the observed image, $f(x, y)$ is the true scene, and $\eta(x, y)$ represents haze.

The key idea of DOS (Dark Object Subtraction) is that some pixels in a scene should naturally be very dark (water bodies, deep shadows). If these pixels show brightness above zero, that added value is most likely haze. We estimate this haze level from the histogram and subtract it from the entire image, which increases contrast and brings the pixel values closer to the true reflectance.

3. Homomorphic Filtering (Illumination Correction)

Uneven lighting is modeled as:

$$f(x, y) = i(x, y) r(x, y)$$

where $i(x, y)$ is illumination (smooth, low-frequency) and $r(x, y)$ is reflectance (sharp, high-frequency).

To separate these components:

1. Take the logarithm: $\ln f = \ln i + \ln r$
2. Apply the DFT: $Z(u, v) = \mathcal{F}\{\ln f(x, y)\}$
3. Use a Gaussian high-pass filter:

$$H(u, v) = (\gamma_H - \gamma_L) \left[1 - e^{-cD^2(u, v)/D_0^2} \right] + \gamma_L$$

4. Perform inverse DFT and exponential transformation.

This suppresses illumination variations and enhances reflectance, helping vegetation features stand out more clearly.

4. Adaptive Spatial Filtering (Noise Removal)

Instead of a simple mean filter, we use the Adaptive Local Noise Reduction Filter, which adjusts its behavior based on local variance:

$$\hat{f}(x, y) = g(x, y) - \frac{\sigma_\eta^2}{\sigma_L^2} [g(x, y) - m_L]$$

If local variance is high (edges), the filter preserves detail. If local variance is low (flat regions), it smooths noise. This makes it especially suitable for satellite imagery.

5. Feature Extraction and Segmentation

We compute three key features:

- **NDVI:** $\frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$ — vegetation density.
- **NDMI:** $\frac{\text{NIR} - \text{SWIR}}{\text{NIR} + \text{SWIR}}$ — vegetation moisture.
- **Hue Angle:** calculated using the HSI formulas from the course book.

These features are stacked and fed into a K-means clustering algorithm. To improve consistency, we initialize the cluster centers using extreme NDVI/NDMI values instead of random seeds.

Time Schedule (5 Weeks)

- **Week 1: Data Acquisition & Initial Checks**

Download satellite images, inspect cloud cover, load Red/NIR/SWIR bands, and prepare the coding environment. Perform basic visual checks to ensure the dataset is appropriate.

- **Week 2: Atmospheric Correction + Class Presentation**

Implement Dark Object Subtraction and show a small presentation in class comparing the raw image vs. haze-corrected output. This presentation will prove that the pipeline is working and justify the next steps.

- **Week 3: Illumination Correction & Noise Reduction**

Implement Homomorphic Filtering (frequency-domain) and Adaptive Noise Reduction. By the end of this week, we aim to have a fully restored image ready for feature extraction.

- **Week 4: NDVI/NDMI/Hue Computation + K-means Segmentation**

Compute the spectral indices, normalize features, and run K-means clustering with deterministic seeds. Begin extracting preliminary vegetation maps.

- **Week 5: Temporal Analysis & Final Report**

Compare vegetation maps across different years, calculate percentage changes, generate visualizations, and finalize the written report and presentation.